Exp. 1 Study of Microsoft Azure and Cloud Service Models

AIM:

To study Microsoft Azure, its major features, cloud service models (IaaS, PaaS, SaaS), and compare different cloud platforms.

THEORY:

i) What is Azure?

Microsoft Azure is a cloud computing platform developed by Microsoft that provides services such as computing, storage, networking, and analytics through the internet. It helps users build, deploy, and manage applications without maintaining physical hardware.

ii) Major Features of Azure

- **Scalability:** Resources can be scaled up or down easily
- **High Availability:** Ensures minimal downtime
- **Security:** Provides encryption and secure access
- **Global Presence:** Data centers across multiple regions
- **Cost Efficiency:** Pay-as-you-go pricing
- **Integration:** Works with Microsoft tools and services
- **Backup & Recovery:** Supports data backup and disaster recovery

iii) Cloud Service Models

1. IaaS (Infrastructure as a Service)

- Provides virtualized computing resources
- User manages OS and applications
- Example: Azure Virtual Machines

2. PaaS (Platform as a Service)

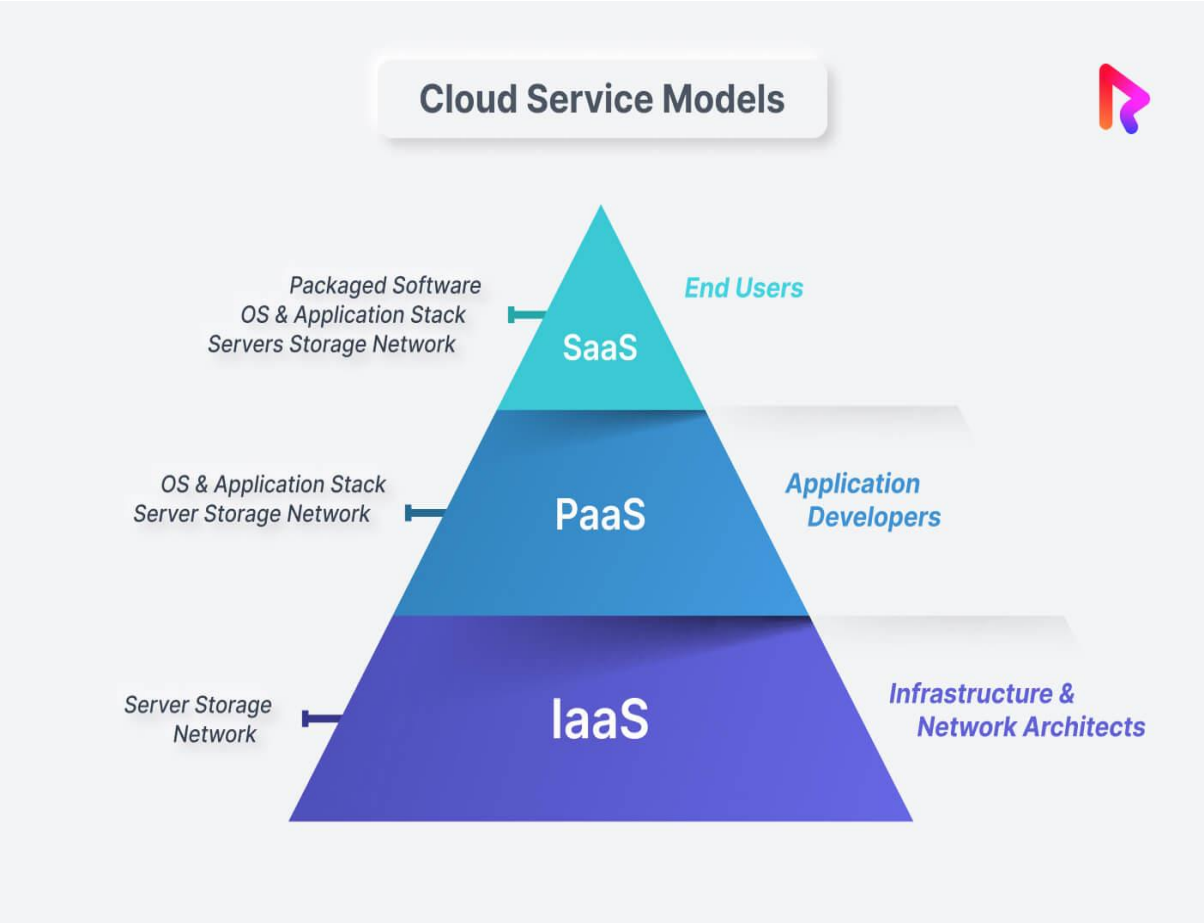
- Provides platform for application development
- No need to manage infrastructure
- Example: Azure App Service

3. SaaS (Software as a Service)

- Software delivered via internet
- No installation required
- Example: Microsoft 365

Comparison of Service Models

Model	Description	User Control	Example
IaaS	Infrastructure	High	Virtual Machines
PaaS	Platform	Medium	App Service
SaaS	Software	Low	Microsoft 365



iv) Cloud Platform Comparison

Comparison between:

- Microsoft Azure
- Amazon Web Services
- Google Cloud Platform

Table Comparison

Feature	Azure	AWS	GCP
Provider	Microsoft	Amazon	Google
Strength	Enterprise integration	Market leader	AI & ML
Pricing	Moderate	Complex	Competitive
Usage	Enterprises	Startups & Enterprises	Data analytics

Compute Services	Elastic Compute Cloud (EC2)	Virtual Machines	Compute Engine
Object Storage	Amazon S3	Azure Blob Storage	Cloud Storage
Networking	Amazon VPC	Azure Virtual Network	Cloud Virtual Network

Microsoft Azure

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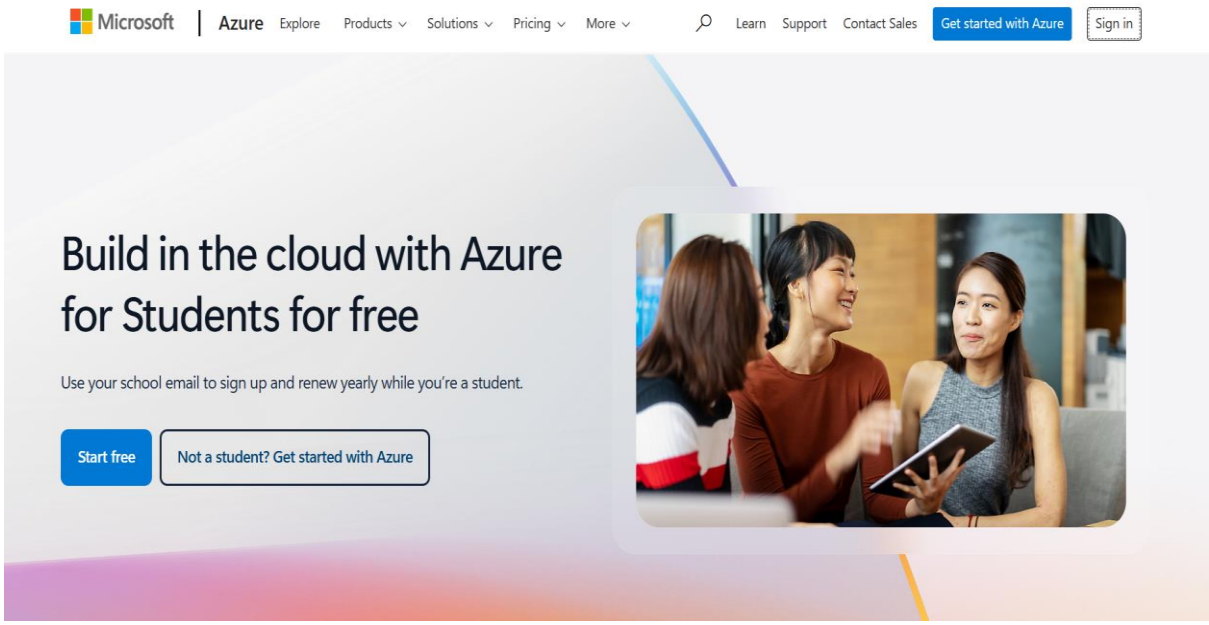
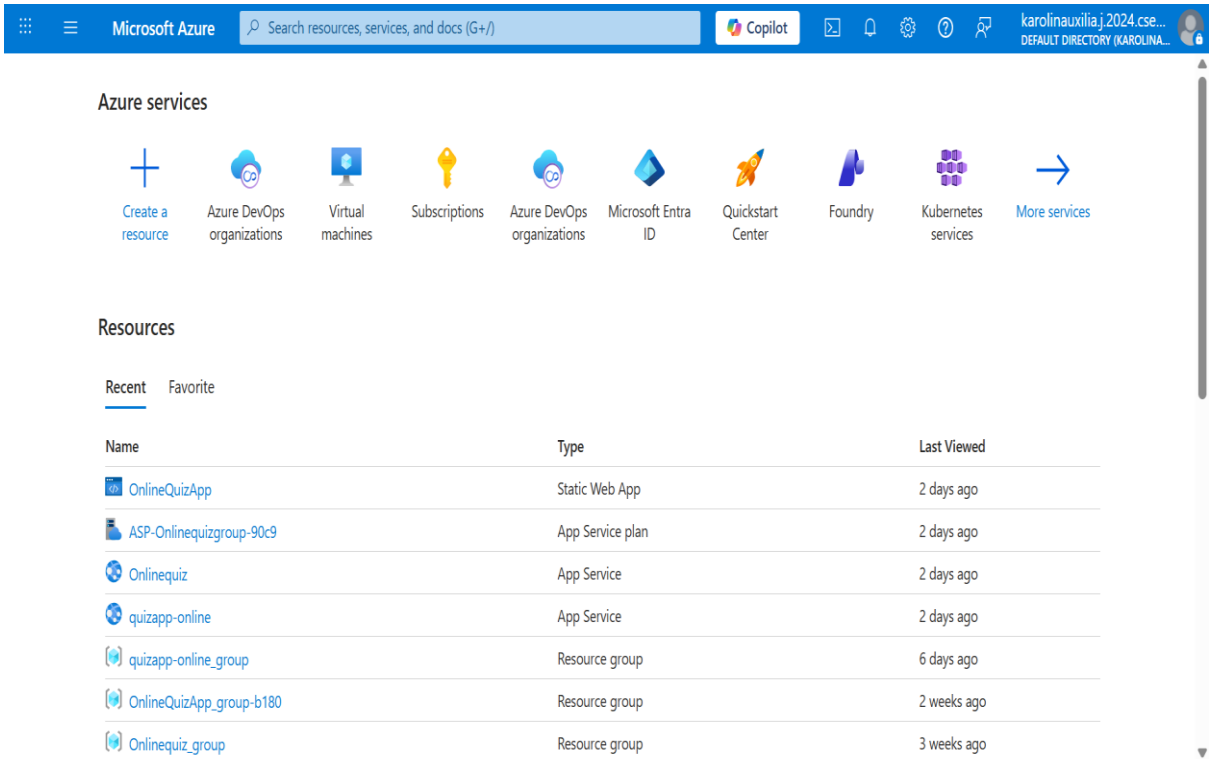
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